

Jingrui Feng

+1-908-400-7046 | fengjingrui@hotmail.com | <https://www.linkedin.com/in/jingrui-feng/> |
Troy, NY 12180, USA

EDUCATION

• Rensselaer Polytechnic Institute

2021 - 2025

B.S. in Information Technology and Web Science

Troy, USA

◦ GPA: 3.57 / 4.00

◦ Relevant Course Work: Data Structure, Intro to Algorithms, Operating Systems, Web System Development, Network Programming, Foundation of Computer Science, Intro Artificial Intelligence, Parallel Programming, Computer Organization, Statistical Methods, Database System, Management IT Resources

EXPERIENCE

• Rensselaer Polytechnic Institute

Jan 2024 - May 2024

Undergraduate Teaching Assistant Operating Systems

Troy, NY

- Provided supports to over 30 students on key operating system concepts such as CPU scheduling, process synchronization, memory management, and basic network programming.
- Held 2 office hours weekly and group sessions to assist with assignments and reinforce understanding of computer architecture and system calls.
- Supported the professor by grading assignments and exams, ensuring timely and fair evaluations of student work.

• Haier Group Corporation

Sep 2023 - Dec 2023

Software Engineer Intern

Qingdao, China

- Conducted testing on the current platform to ensure reliable connectivity and system stability, focusing on performance across multiple environments.
- Learned and applied Flutter technology, enabling seamless deployment of the software across multiple platforms, including Linux, Windows, and macOS.
- Improved communication and collaboration with overseas clients, ensuring alignment on project goals and technical requirements.
- Gained expertise in software architecture, identifying optimization opportunities to enhance system performance and streamline the development process.

• NASA Renewable Energy Data Visualization

Jan 2023 - Jul 2023

Full-Stack Developer

Troy, NY

- Leveraged NASA's Prediction Of Worldwide Energy Resource (POWER) data to evaluate solar and wind energy potential across counties in the United States, providing detailed insights into regional renewable energy viability.
- Developed an interactive data visualization platform using React, enabling users to explore energy consumption trends and identify optimal renewable energy sources through color-coded mapping and dynamic filtering options.
- Engineered a high-performance, stateless REST API with Node.js and MongoDB to ensure rapid data retrieval and efficient back-end operations.
- Presented research findings at the July 2023 Earth Science Information Partners (ESIP) Meeting, showcasing the platform's ability to enhance public understanding of renewable energy data.

RESEARCH

• Chinese Academy of Sciences

Jul 2024 - Sep 2024

Research Assistant in Machine Learning

Remote

- Gained foundational experience in machine learning by studying key research papers and learning algorithms such as the Transformer model and other machine learning techniques.
- Worked under the guidance of a research professor, contributing to small-scale machine learning projects to apply theoretical knowledge in practical scenarios.

PROJECTS

• MissionFlow - United Airline Airshop Lab

Sep 2024 - Dec 2024

Frontend Developer

Troy, NY

- Developed MissionFlow, a comprehensive web-based application in collaboration with United Airlines' Airshop Lab, addressing inefficiencies in managing robotic R&D missions and improving operational scalability and innovation.
- Streamlined mission file management and execution by providing a centralized system for organizing mission data, automating file transfer, command execution, and post-mission data handling, reducing manual overhead by 50%.
- Integrated advanced analytics and API compatibility with Boston Dynamics systems, enabling real-time insights and seamless robotic platform management, while reducing error rates by 15%.
- Enhanced operational efficiency and scalability, increasing the capacity to execute complex robotic scenarios and reinforcing United Airlines' leadership in robotics and aviation innovation.

• Ryan's Ice Cream Online Ordering System

Jan 2024 - Apr 2024

Developer

Troy, NY

- Designed and developed an online cake-ordering system using AppFront, modernizing business operations and delivering a seamless, intuitive user experience for customers to browse, customize, and purchase cakes, contributing to a significant increase in overall project benefits.
- Integrated secure payment processing, real-time order tracking, and inventory management, resulting in a 20% rate of return and ensuring efficient order fulfillment, boosting operational efficiency while minimizing project costs.
- Implemented dynamic website updates, including optimization and community-driven features, enabling real-time customer engagement and brand interaction, which contributed to substantial growth in future business value.
- Delivered a robust back-end solution for managing orders, analyzing customer data, and facilitating future business growth and technological enhancements, reinforcing the profitability and viability of the project.

• RPI Foodies

Sep 2022 - Jan 2023

Full-Stack Developer

Troy, NY

- Designed and developed an online platform to enhance RPI students' dining experience by allowing users to post and share food images from campus dining halls, enabling community engagement through comments and likes.
- Implemented a ranking system to highlight top food posts weekly based on user engagement, offering insights into the most popular dining halls.
- Developed a comprehensive admin system with content moderation and reporting features to ensure the platform remains secure and free from spam or inappropriate content.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, HTML, CSS, C++, C, Java
- **Web Technologies:** React, Node.js, Flutter
- **Database Systems:** MySQL, MongoDB, PostgreSQL
- **Others:** Agile, Waterfall, Excel, Git, RESTful API